From Shor's Threat to Kyber's Shield: Lattice Cryptography from Scratch

Toy parameters are for learning only, never for real data.

1. Problem: Why Replace RSA?

- **Shor's algorithm** breaks RSA and ECC on a large quantum computer.
- **LWE** hides a secret $\mathbf{s}$ in noisy equations $\mathbf{b} = A\mathbf{s}+\mathbf{e} \pmod q$. It has no periodic structure for a QFT to exploit, so it underpins NIST's ML-KEM (Kyber).
- **Our goal:** toy LWE (n=32, m=64, q=3329), two attacks, a timing-leak fix, Ring-LWE with NTT, and Shor for N=15.

2. Toy LWE & Noise

- **Decrypt:** center $d = v - \mathbf{s}^T\mathbf{u}$ into $(-q/2, q/2]$; bit is 0 if $|d| < q/4$.
- $B = 0$ **is insecure:** Gaussian elimination recovers $\mathbf{s}$ in **9.4 ms** (100%). For $B \geq 1$ recovery drops to **0%**.
- **Noise trade-off:** $\sigma^2_{\text{total}} = \frac{mB(B+1)}{6}$, $P_{\text{fail}} \approx 2\big(1 - \Phi(\frac{q/4}{\sigma_{\text{total}}})\big)$.

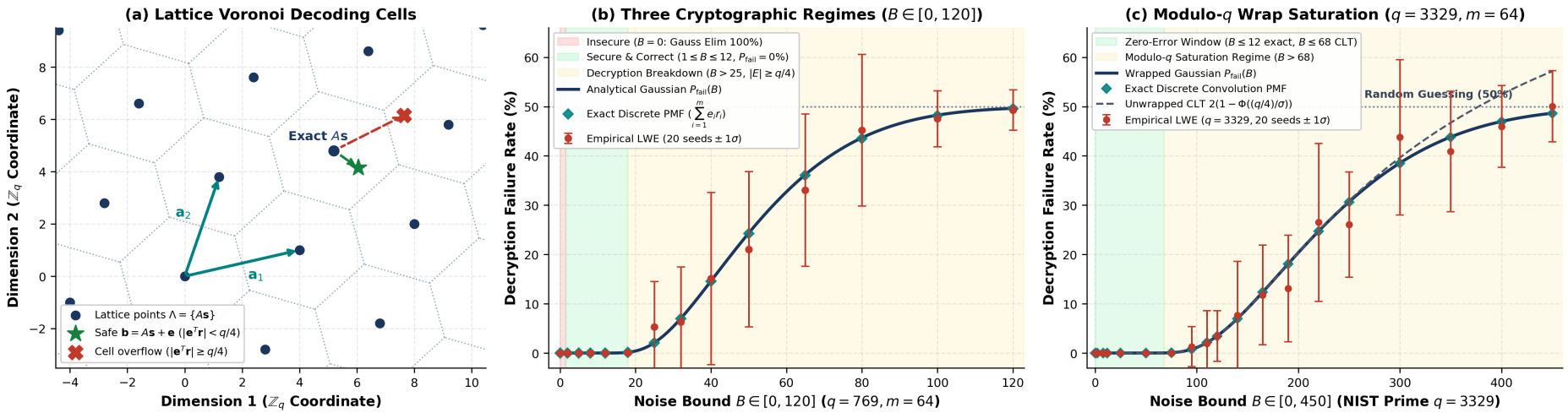


Figure 1: (a) small noise keeps $\mathbf{b}$ in its cell, large noise crosses the boundary. (b, c) decryption failure vs. noise bound B matches our formula.

3. Breaking Toy LWE: Brute Force vs. LLL

- **Brute force** $O(3^n)$: 100% up to n=11 (102 ms), but **212.6 s** at n=18 and impossible at $n \geq 32$.
- **Kannan embedding + LLL:** turns LWE into a short-vector problem, since $(\mathbf{e}, -\mathbf{s}, 1)$ is unusually short. 100% recovery for $n \leq 8$, but **0% at** $n \geq 16$.

4. Timing Leak & Fix

- **Leak:** an early-exit check leaks **1.85** μ**s** per matching coefficient ($R^2 = 0.9952$).
- **Exploit:** a timing oracle recovers a 16-coefficient secret in **48 queries** instead of $3^{16} \approx 43$ million.
- **Fix:** a constant-time XOR accumulator gives a flat runtime (452.4 $\pm$ 31.1 μs).

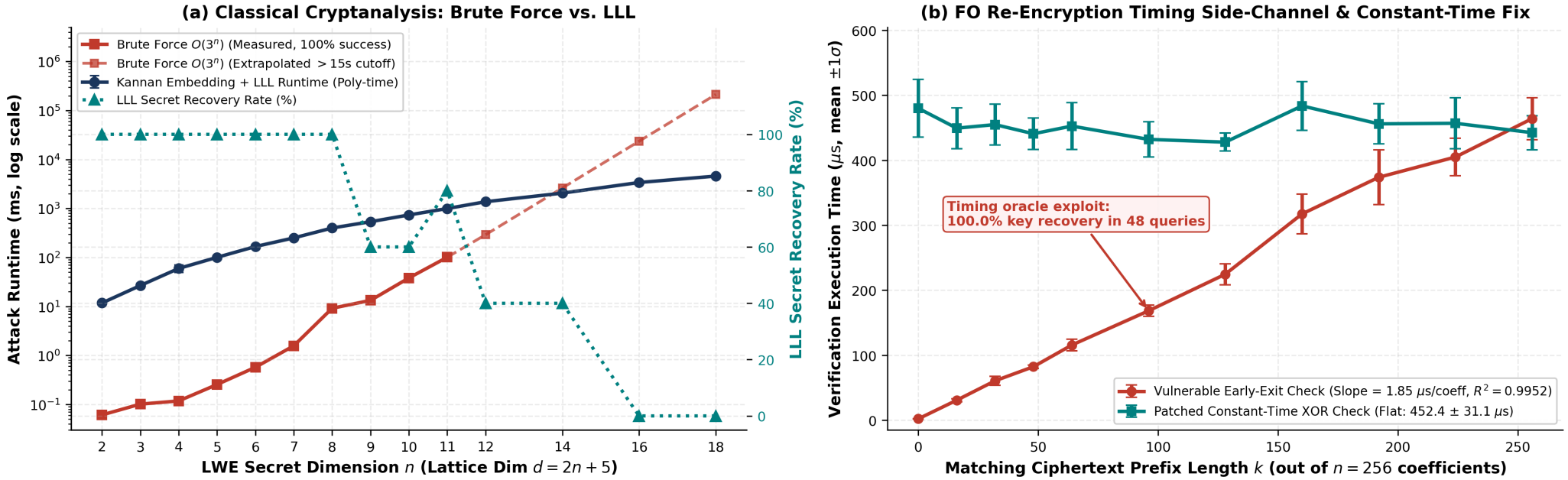


Figure 2: (a) brute force vs. LLL runtime and LLL success rate. (b) early-exit time grows with prefix length; XOR check stays flat.

5. Shor's Circuit for $N = 15$

- Pick $a = 7$: the period of $7^x \bmod 15$ is $r = 4$, so $\gcd(7^2 \pm 1, 15) = \{3, 5\}$.
- **5-qubit circuit:** 3 counting + 2 work qubits, 11 two-qubit gates.
- **Results (4,096 shots):** ideal simulator 100% on the correct peaks; IBM Eagle noise model **91.9%**, still giving $r = 4$.

6. Ring-LWE & NTT

- **Smaller keys:** polynomials in $\mathbb{Z}_q[x]/(x^n + 1)$ shrink the public key by over **1000**× (768 bytes).
- **Faster:** NTT multiplies in $O(n \log n)$ vs. $O(n^2)$, **15.7**× faster at n=1024 (0.74 ms vs. 11.62 ms).
- Our 256-bit Kyber-style KEM has **0.0%** bit error rate.

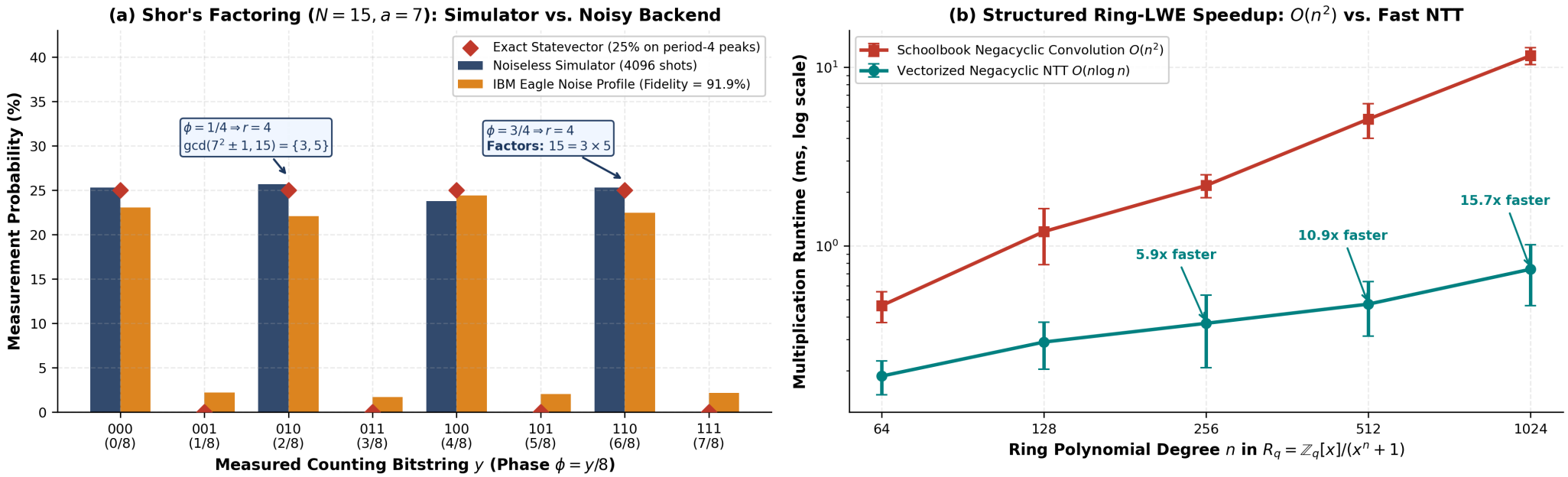


Figure 3: (a) Shor phase histogram, simulator vs. noisy IBM Eagle. (b) NTT vs. schoolbook polynomial multiplication.

Key & Ciphertext Sizes (bytes)

Scheme	PK	CT	Quantum-safe?
RSA-2048	256	256	No
ECC P-256	64	64	No
Toy LWE (n=32)	3,168	12,672	No (toy)
Standard LWE (n=512)	787,968	196,992	Yes
Ring-LWE (n=256)	768	768	Yes
ML-KEM-768	1,184	1,088	Yes

Quantum Resources to Break Keys

Target	Logical Q	Physical Q	Works today?
Shor N=15 (ours)	5	5	Yes (91.9%)
RSA-2048	4,099	∼20M	No
ECC P-256	2,330	∼6M	No
ML-KEM-768	N/A	N/A	Immune

Estimates from Gidney & Ekerå (2021).

7. Honest Limits

- LLL breaks tiny toy sizes ($n \leq 8$) but fails at $n \geq 16$. Real Kyber is infeasible ($> 2^{140}$ operations).
- The Shor circuit is hand-compiled for N=15. A general circuit needs far more qubits and error correction.
- Toy only: no Module-LWE, centered binomial sampling or FO hashing. Never use for real data.

8. AI-Use Note: What the AI Got Wrong

- An AI assistant helped draft structure and plots; 8 unit tests verified every formula and circuit.
- **Bug:** the AI's `(v - s.u) % q` skipped centering, so a small negative noise like -3 became 3326 and bit 0 decoded as 1 (**15.85%** error at B=1).
- **Fix:** center into $(-q/2, q/2]$, giving **0.00%** error over 2,000 trials.